Souvik Roy

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SUMMARY

Results-driven Machine Learning and Full Stack Developer with experience building and deploying end-to-end ML systems and AI-powered web applications. Skilled in Scikit-Learn, FastAPI, React, and Node.js with a focus on model training, evaluation, and real-world deployment. Proficient in integrating ML pipelines into production full-stack products.

SKILLS

Machine Learning:	Scikit-Learn, Feature Engineering, Hyperparameter Tuning, Model Evaluation, Logistic Regression, Random Forest, XGBoost, TensorFlow, Deep Learning
Programming:	Python, JavaScript, Object-Oriented Programming
Frontend:	React, HTML, CSS
Backend:	Node.js, Express.js, FastAPI
Database:	MongoDB, SQL
Tools:	Git, GitHub, REST APIs, Vercel, Render, TensorFlow.js

EDUCATION

Bachelor of Engineering - Civil Engineering — Nov 2024 - Present
Jadavpur University • Kolkata, West Bengal

Higher Secondary Education - Science (PCM) — May 2024
St. Patrick's Higher Secondary School • Asansol, West Bengal

EXPERIENCE

ET GenAI Hackathon — AI Track Participant — Mar 2026
Economic Times of India • Track 8: AI-Native News Experience

- Built a 3-agent autonomous AI pipeline using Groq (Llama models) for real-time news ingestion, story arc classification, and insight generation.
- Integrated Tavily Search API for live data retrieval and designed intelligent model routing to optimise inference latency across agents.
- Deployed a full-stack React and Node.js application with multi-turn conversational AI; awarded Top Team recognition by the Economic Times.

PROJECTS

IPL Win Predictor | Python, Scikit-Learn, FastAPI, React — May 2026 - Jun 2026 | Code | Live Demo

- Built a full-stack ML application predicting live IPL second-innings win probability with an interactive React dashboard.
- Engineered match-state features (score, wickets, overs, run-rate, required run-rate) and benchmarked Logistic Regression, Random Forest, and XGBoost.
- Achieved ~82.6% accuracy and 0.91 ROC-AUC via hyperparameter tuning and cross-validation; deployed FastAPI backend to Render and React frontend to Vercel.

CityPulse | JavaScript, TensorFlow 2.0, React, Node.js, MongoDB, Groq — Mar 2026 - May 2026 | Code | Live Demo

- Built ML dataset generation pipelines for AQI and weather data; trained TensorFlow models to predict next-hour AQI and weather conditions for Indian cities.
- Integrated AI-powered AQI analysis using Groq SDK (Llama models) to deliver natural-language environmental insights alongside predictive model output.
- Deployed end-to-end full-stack platform on Vercel and Render featuring geospatial search and interactive map visualisation via OpenStreetMap APIs.

CERTIFICATIONS

- ET GenAI Hackathon Top Team Certificate - Economic Times of India, Issued 2026
- Complete Data Science Course by Daniel Bourke - Udemy, Issued 2026
- AI Tools and ChatGPT Workshop - be10X, Issued 2025